

AWS Identity and Access Management (IAM)

Assessment no: 01

Comprehensive Guide with Hands-On Steps and Visuals

Overview

AWS Identity and Access Management (IAM) is a web service that helps you securely control access to AWS resources.

- **Authentication:** Who can sign in.
- **Authorization:** What they can do.
- **Benefits:** Fine-grained access control, centralized management, and improved security by enforcing the principle of least privilege.

IAM is built on four key components:

1. **Users**
2. **Groups**
3. **Policies**
4. **Roles**

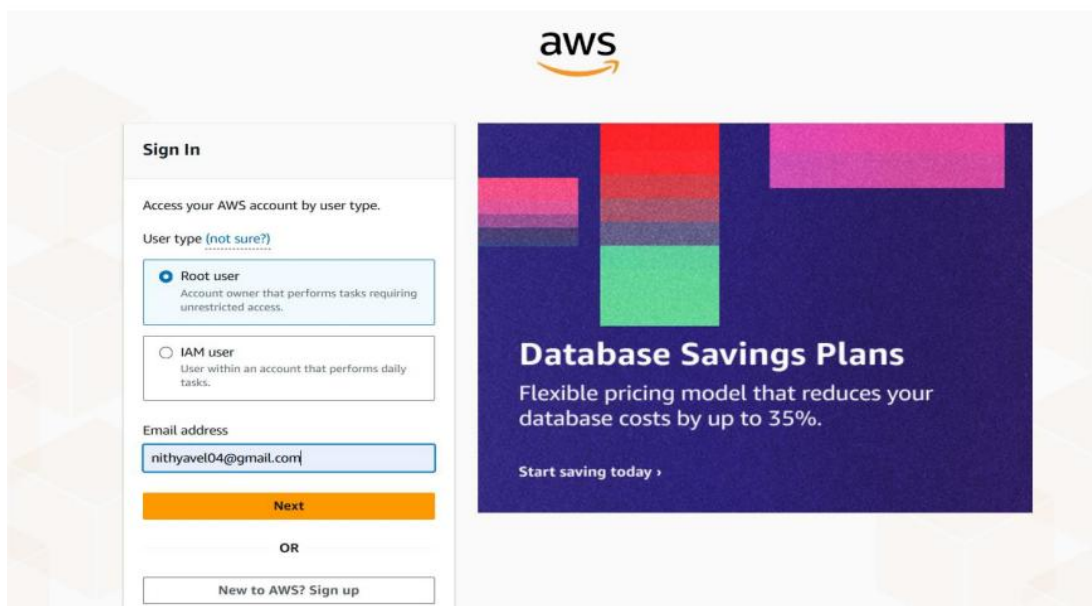
1. IAM Users

An **IAM User** is an individual identity created in IAM with specific credentials (username, password, access keys).

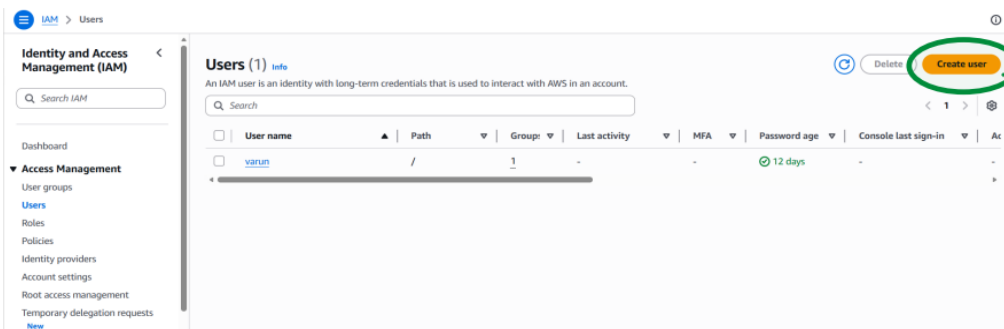
- Example: A developer with access to EC2 instances.
- Best practice: Avoid using the root account for daily tasks; create IAM users instead.

Hands-On: Creating an IAM User (*Nithya*)

1. **Login as Root User**



2. Navigate to IAM → Users → Create User



3. Specify User Details

- Username: *Nithya*
- Console access enabled
- Autogenerated password
- Enforce password reset at next sign-in

Specify user details

User details

User name

Nithya

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, -, ., @, _ (hyphen)

☒ **Provide user access to the AWS Management Console - optional**

In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☒ **Autogenerated password**

You can view the password after you create the user.

☐ **Custom password**

Enter a custom password for the user:

Must be at least 8 characters long
Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } ' "

☐ Show password

☒ **Users must create a new password at next sign-in - Recommended**

Users automatically get the `AWSCognitoChangePassword` policy to allow them to change their own password.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.

[Learn more](#)

Cancel Next

4. Set Permissions

- Attach **IAMUserChangePassword** policy

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☒ **Add user to group**

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ **Attach policies directly**

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1)

Search

Group name	Users	Attached policies	Created
developer	0	AmazonEC2ContainerRegistryFullAccess	2026-02-27 (17 days ago)

Set permissions boundary - optional

Cancel Previous Next

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name: Nithya
Console password type: Autogenerated
Require password reset: Yes

Permissions summary

Name	Type	Used as
IAMUserChangePassword	AWS managed	Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

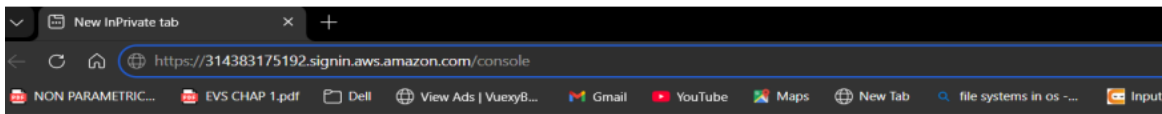
Cancel Previous Create user

5. Review and Create User

6. Download Credentials (CSV)

7. Login as IAM User

- Use console link from CSV



InPrivate browsing

- Enter username and password

IAM user sign in ⓘ

Account ID or alias (Don't have?)
314383175192

☐ Remember this account

IAM username
Nithya

Password

☐ Show Password [Having trouble?](#)

Sign in

- Reset password at first login

Password reset ⓘ

Your account (314383175192) password has expired or requires a reset.
To continue, please verify your old and set a new password for Nithya (not you?).

Old Password

☐ Show Password

New Password

Confirm New Password

☐ Show Password **Matches**

Confirm Password Change

[Sign in to a different account](#)

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Password Reset: If a user forgets their password, the root user can reset it via IAM console under *Manage console access*.

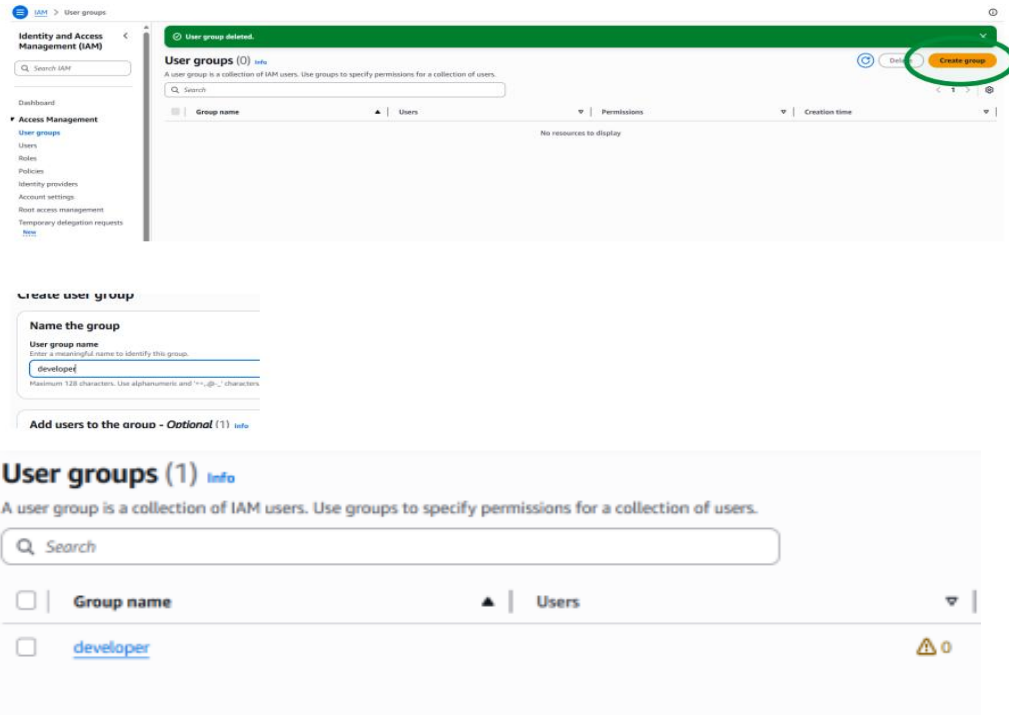
2. IAM Groups

An **IAM Group** is a collection of users with shared permissions.

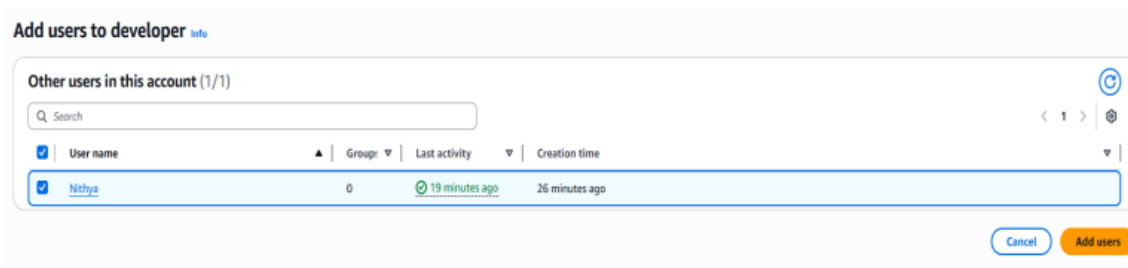
- Benefit: Easier to manage permissions at group level instead of individual users.
- Example: Create a group named *developer* and add all developer users to it.

Hands-On: Creating a Group (*developer*)

1. Go to IAM → User Groups → Create Group
2. Name the Group (*developer*)



3. Add Users (e.g., Nithya)



3. IAM Policies

An **IAM Policy** is a JSON document that defines permissions. Policies are the backbone of IAM security. They specify what actions are allowed or denied on AWS resources.

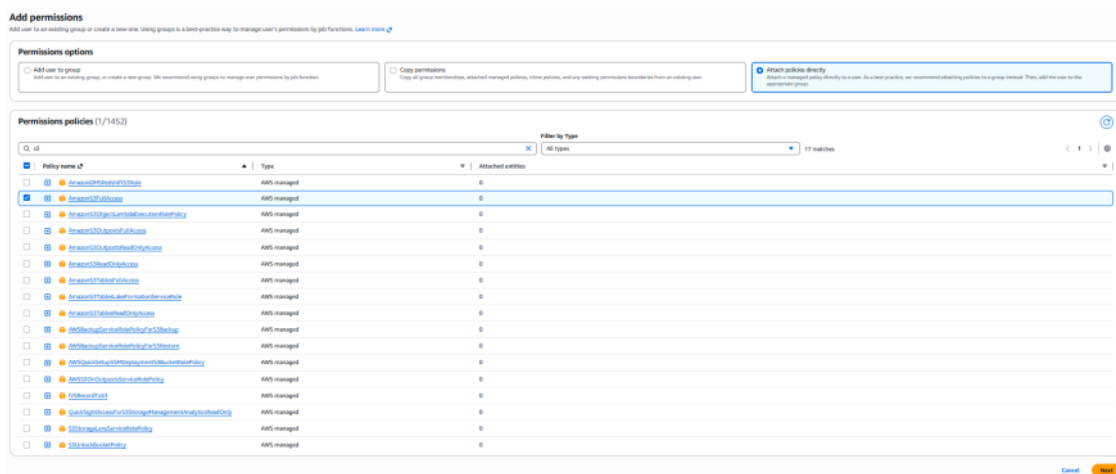
Types of Policies

1. **Identity-based policies** (attached to users, groups, or roles)
 - **Managed policies:**
 - *AWS Managed Policies:* Predefined by AWS (e.g., AmazonS3FullAccess).
 - *Customer Managed Policies:* Custom policies created by the user.
 - **Inline policies:** Embedded directly into a single IAM identity (user/group/role). One-to-one relationship.

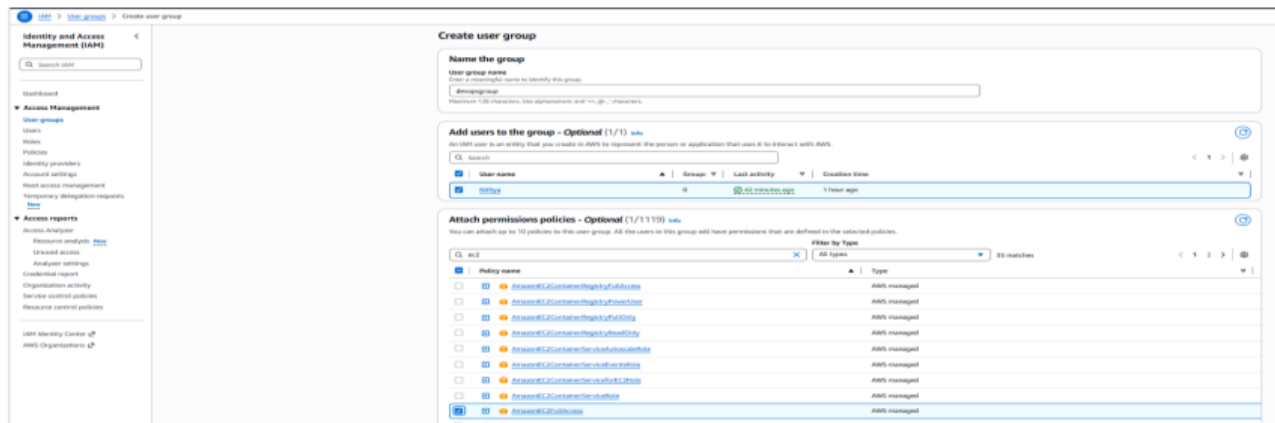
2. **Resource-based policies** (attached directly to resources like S3 buckets, Lambda functions, SNS topics).
 - Useful for cross-account access.

Hands-On Examples

- **Attach AmazonS3FullAccess directly to user Nithya**



- **Create a group devopsgroup, add Nithya, and attach AmazonEC2FullAccess**



- **Create a custom policy for specific needs and attach it to a user or group**



4. IAM Roles

An **IAM Role** is an identity with temporary credentials that can be assumed by users, applications, or AWS services.

- Unlike users, roles are not tied to a specific person.
- Roles are ideal for service-to-service communication or temporary access.

Use Cases

- **Service-based:** EC2 instance assuming a role to access S3.
- **Account-based:** Temporary access for another AWS account.

Accessing a Role via STS (Security Token Service)

- Add an inline STS policy to the user (e.g., *Nithya*) to allow role assumption.

Nithya [info](#) [Delete](#)

Summary

ARN: `arn:aws:iam::31435175192:user/Nithya`

Created: March 12, 2026, 18:59 (UTC+05:30)

Console access: Enabled without MFA

Last console sign-in: Never

Access key 1: [Create access key](#)

Permissions | Groups | Tags (1) | Security credentials | Last Accessed

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

Policy name: [IAMUserChangePassword](#) | Type: AWS managed | Attached via: Directly

Permissions boundary (not set)

Generate policy based on CloudTrail events

STS [info](#) [Add permissions](#)

Actions allowed

Specify what actions can be performed on specific resources in STS.

Filter Actions

Manual actions | [Add actions](#)

☒ All STS actions (sts:*)

Access level

Read (Selected 5/5)

Write (Selected 9/9)

Tagging (Selected 2/2)

Resources

Specify resource ARNs for these actions.

☒ All

☐ Specific

The all wildcard "*" may be overly permissive for the selected actions. Allowing specific ARNs for these service resources can improve security.

Request conditions - optional

Actions on resources are allowed or denied only when these conditions are met.

[Add more permissions](#)

Security: 0 | Errors: 0 | Warnings: 0 | Suggestions: 0

[Cancel](#)

Review and create [info](#)

Review the permissions, specify details, and tags.

Policy details

Policy name:

Maximum 128 characters. Use alphanumeric and "+, -, @, _" characters.

Permissions defined in this policy [info](#) [Edit](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, group, or role), attach a policy to it.

Search

Allow (1 of 464 services) [Show remaining 463 services](#)

Service	Access level	Resource	Request condition
STS	Full access	All resources	None

[Cancel](#) [Previous](#) [Create policy](#)

- Use *Switch Role* in the AWS console to assume *devopsrole1*.

Console Home [info](#) [Reset to default layout](#)

Recently visited [info](#)

No recently visited services

Explore one of these commonly visited AWS services.

[EC2](#) [S3](#) [CloudWatch](#) [IAM](#)

[View all services](#)

Applications (0) [info](#)

Region: Europe (Stockholm)

Select Region: eu-north-1 (Current Region) [Find applications](#)

Name	Description	Region
Access denied to servicecatalog:ListApplications		

[Diagnose with Amazon Q](#)

[Go to myApplications](#)

Account ID: 3143-8517-5192

Account name: Access denied

Account color: Access denied

IAM user: Nithya

Account:

- Organization
- Service Quotas
- Billing and Cost Management
- Security credentials
- Console Mobile App

[Turn on multi-session support](#)

[Switch role](#) [Sign out](#)

Switch Role

Switching roles enables you to manage resources across Amazon Web Services. You can temporarily take on the permissions assigned to the new role. When you are done, you can return your permissions back. [Learn more](#)

Account ID

The 12-digit account number or the alias of the account in which the role is located.

314383175192

IAM role name

The name of the role that you want to assume which can be found at the end of the role name from the following role ARN: `arn:aws:iam::123456789012:role/TestRole`

devopsrole1

Display name - optional

This name will appear in the console navigation bar when active. Choose a name that is unique and descriptive.

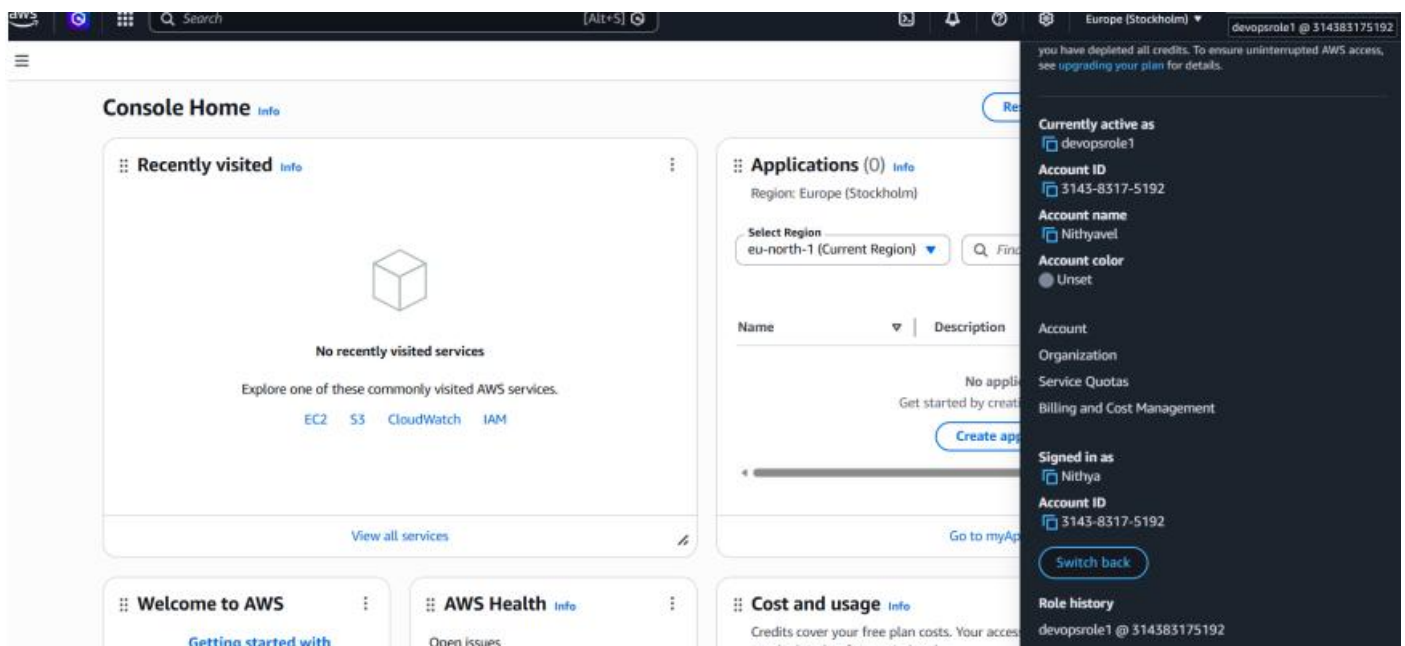
devopsrole1 @ 314383175192

Display color - optional

The selected color displays in the console navigation when this role is active.

☐ None

- Once switched, the user temporarily gains the permissions of the role.



Best Practices

- Always enforce **least privilege**.
- Use **groups** for permission management instead of attaching policies directly to users.
- Rotate and manage **access keys** regularly.
- Prefer **roles** for temporary access and service-to-service communication.